Module B6 — Biotechnology, Ecosystems and Human Impact

6.1 Gene Technology

Polymerase Chain Reaction (PCR)

PCR is a technique that amplifies (copies) a specific fragment of DNA millions of times in vitro. It requires a DNA template, DNA primers (short complementary sequences that flank the target region), free nucleotides, and a thermostable DNA polymerase (Taq polymerase, from Thermus aquaticus bacteria that live in hot springs).

Each PCR cycle involves three temperature-controlled steps:

1. **Denaturation (94-98 degrees C):** Heat breaks hydrogen bonds between complementary base pairs, separating the double-stranded DNA into two single strands
2. **Annealing (50-65 degrees C):** Temperature is lowered to allow primers to bind (anneal) to their complementary sequences on the template strands by hydrogen bonding
3. **Extension (72 degrees C):** Taq polymerase extends from the primers, adding free nucleotides complementary to the template strand, synthesising new DNA strands

After n cycles, approximately 2^n copies of the target DNA are produced. After 30 cycles, this gives over one billion copies from a single starting molecule. PCR is used in forensic science (DNA profiling from tiny samples), medical diagnosis (detecting pathogen DNA), genetic research, and paternity testing.

Gel Electrophoresis

Gel electrophoresis separates DNA fragments by size. DNA is cut using restriction enzymes (which cut at specific recognition sequences) or amplified by PCR, then loaded into wells in an agarose gel and an electric current is applied.

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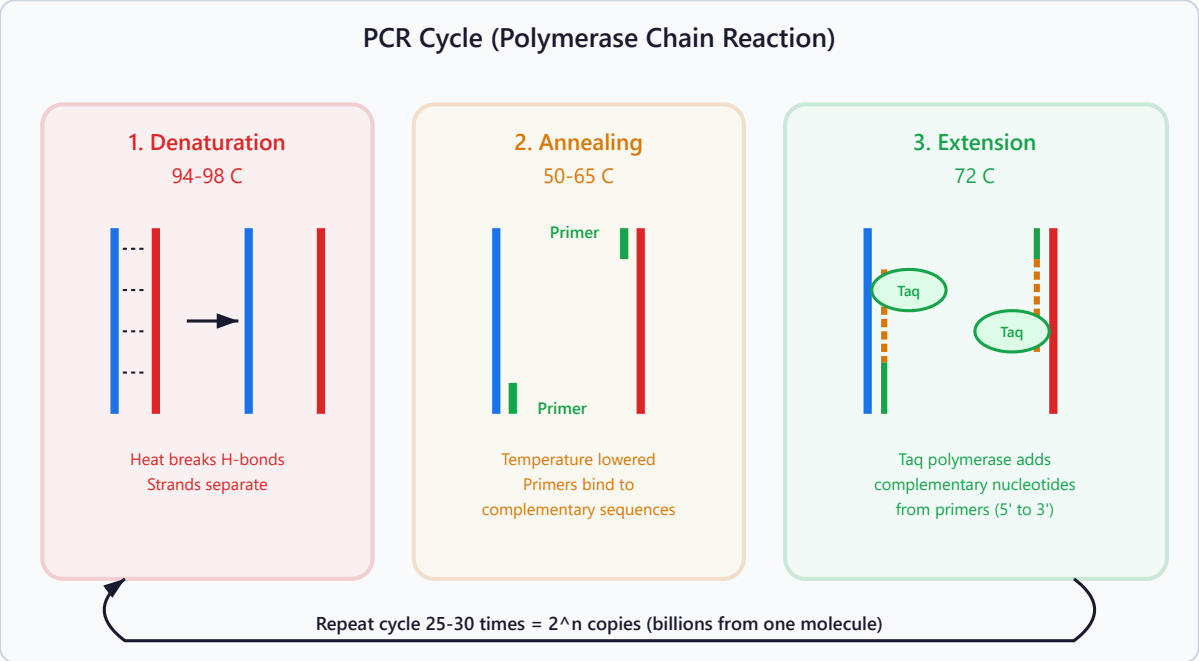
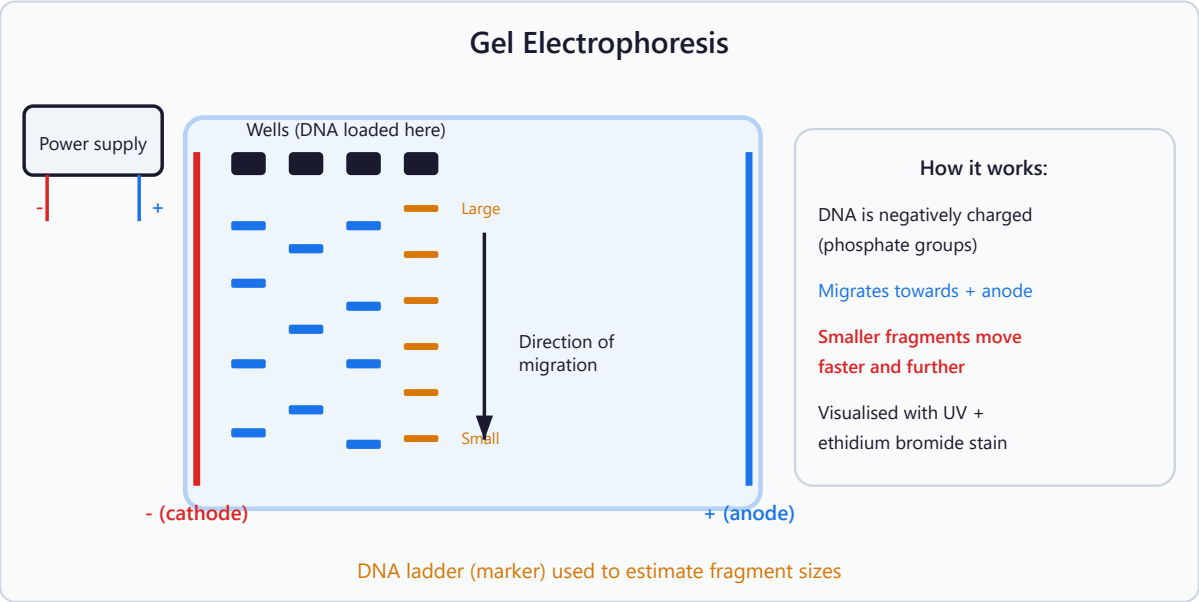
DNA is negatively charged (due to phosphate groups), so fragments migrate towards the positive electrode (anode). Smaller fragments move faster and further through the gel pores. After separation, DNA is visualised using staining (e.g., ethidium bromide under UV light) or radioactive/fluorescent probes.

Restriction enzymes: Bacterial enzymes that cut DNA at specific palindromic recognition sequences. They may produce blunt ends or sticky ends (short single-stranded overhangs).
DNA ligase: An enzyme that joins DNA fragments by forming phosphodiester bonds between adjacent nucleotides. Used to seal recombinant DNA molecules.



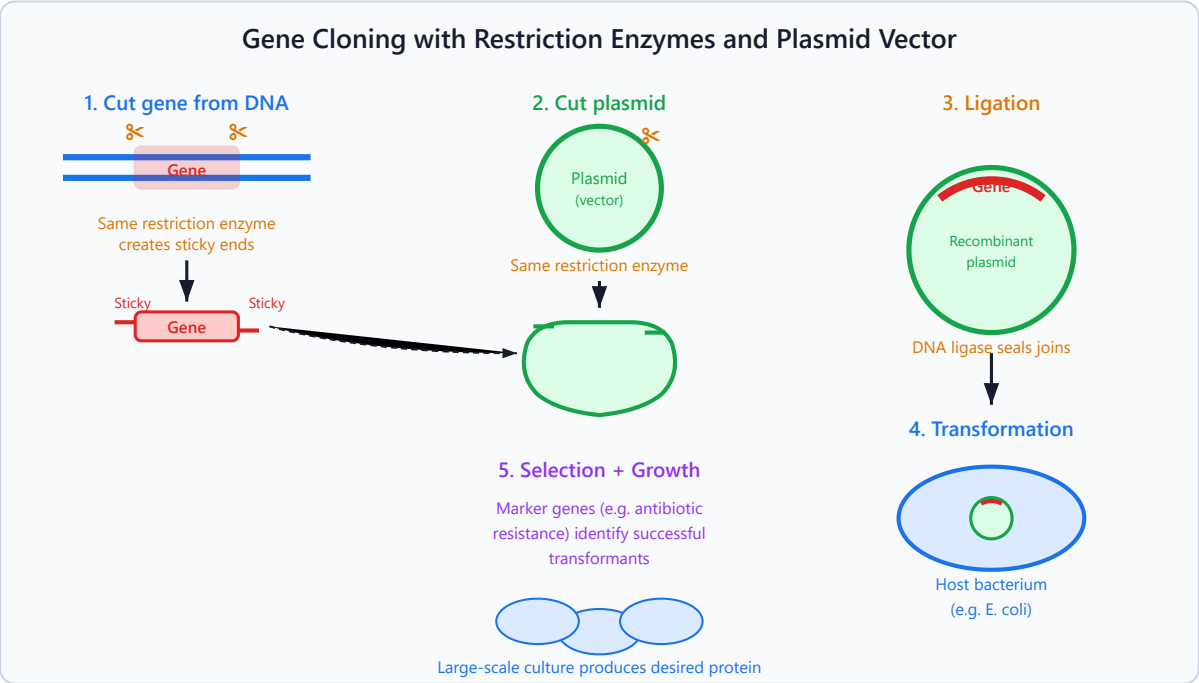


Genetic Engineering (Recombinant DNA Technology)

Genetic engineering involves transferring a gene from one organism to another (often a different species) to produce a desired protein or trait. The key steps are:

1. **Identification and isolation of the gene:** Using restriction enzymes to cut it from the genome, reverse transcriptase to make cDNA from mRNA, or chemical synthesis based on the known amino acid sequence
2. **Insertion into a vector:** The gene is ligated into a vector (plasmid, virus, or liposome) using the same restriction enzyme to create compatible sticky ends, then DNA ligase seals the joins. A promoter sequence is added to ensure the gene is expressed.
3. **Transformation:** The vector is introduced into the host cell (e.g., bacteria, yeast, plant or animal cells) using heat shock, electroporation, or microinjection
4. **Selection:** Identifying which host cells have successfully taken up the recombinant DNA, using marker genes (e.g., antibiotic resistance genes or fluorescent protein genes)
5. **Expression and growth:** Host cells that express the desired gene are cultured on a large scale to produce the protein product

Applications include production of human insulin by genetically modified bacteria, golden rice (containing beta-carotene), Bt crops (containing insecticidal protein from Bacillus thuringiensis), and production of factor VIII for haemophilia treatment.



Gene Therapy

Gene therapy involves inserting a functional copy of a gene into the cells of a patient who has a defective or missing gene, to treat or cure a genetic disorder. It can be:

- **Somatic gene therapy:** Targets body cells of the patient. Effects are not passed to offspring. Examples include treating cystic fibrosis (inserting functional CFTR gene into lung epithelial cells) and severe combined immunodeficiency (SCID).
- **Germ-line gene therapy:** Targets gametes or early embryos. Changes would be inherited by offspring. Currently not permitted in humans due to ethical concerns and safety uncertainties.

Vectors for gene therapy include modified viruses (adenoviruses, retroviruses, lentiviruses) and liposomes. Challenges include the immune system attacking vectors, difficulty targeting the right cells, short-lived effects requiring repeated treatment, and the risk of insertional mutagenesis (gene inserting into an oncogene or tumour suppressor gene).

> **Exam Tip:** Gene therapy replaces or supplements a defective gene; it does not repair the existing faulty gene. The original mutant gene is still present in the DNA. Also remember that somatic gene therapy must often be repeated because the treated cells are eventually replaced by new cells that do not contain the functional gene.

6.2 Cloning

Natural and Artificial Cloning in Plants

Plants can be cloned naturally through vegetative propagation (runners, bulbs, tubers) or artificially through:

- **Cuttings:** Sections of stem are cut and placed in moist soil or rooting hormone; they develop adventitious roots
- **Tissue culture (micropropagation):** Small pieces of plant tissue (explants) are grown on sterile nutrient agar containing growth hormones (auxins and cytokinins). Cells divide to form a callus, which is then subcultured to produce many genetically identical plantlets. Requires aseptic technique throughout.

Animal Cloning

Somatic cell nuclear transfer (SCNT): The nucleus from a somatic (body) cell of the organism to be cloned is transferred into an enucleated egg cell. The egg is stimulated to divide and develops into an embryo, which is implanted into a surrogate mother. Dolly the sheep (1996) was the first mammal cloned from an adult somatic cell.

Applications include preserving endangered species, producing genetically identical animals for research, and replicating organisms with desirable agricultural traits. Limitations include low success rates, premature ageing, and ethical concerns about welfare.

6.3 Biotechnology and Microorganisms

Using Microorganisms in Industry

Microorganisms are widely used in biotechnology because they reproduce rapidly, have simple nutritional requirements, can be genetically modified easily, and produce a wide range of useful products.

Product	Organism	Application
Insulin	Genetically modified E. coli or yeast	Treatment of diabetes
Antibiotics (penicillin)	Penicillium fungi	Treatment of bacterial infections
Bread, beer, wine	Saccharomyces cerevisiae (yeast)	Food and drink production via fermentation
Cheese, yoghurt	Lactobacillus bacteria	Dairy production via lactic acid fermentation
Mycoprotein (Quorn)	Fusarium venenatum	High-protein food source
Biofuels (bioethanol)	Yeast	Renewable energy from fermentation of plant sugars

Fermentation and Bioreactors

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Industrial fermentation takes place in bioreactors (fermenters) that provide optimal conditions for microbial growth. Key features include:

- **Sterile conditions:** The vessel is sterilised before use and air is filtered to prevent contamination by unwanted organisms
- **Temperature control:** Water-cooled jacket removes metabolic heat to maintain optimum temperature for enzyme activity
- **pH control:** Buffers or acid/alkali additions maintain optimum pH
- **Oxygen supply:** Sterile air is bubbled through for aerobic organisms; agitators (stirrers) ensure even distribution of nutrients and oxygen
- **Nutrient supply:** Carbon source, nitrogen source, minerals, and vitamins are provided

Batch culture involves adding all nutrients at the start and harvesting at the end. The population follows a typical growth curve: lag phase (adaptation, enzyme synthesis), log/exponential phase (rapid reproduction, unlimited resources), stationary phase (nutrient depletion, waste accumulation, death rate equals birth rate), and decline/death phase. **Continuous culture** involves constantly adding fresh nutrients and removing product, maintaining the population in the exponential growth phase for maximum productivity.

Feature	Batch Culture	Continuous Culture
Nutrient supply	All added at start	Continuously added
Product harvesting	At end of fermentation	Continuously removed
Growth phase maintained	All phases occur	Exponential phase maintained
Contamination risk	Lower (vessel sealed)	Higher (continuous addition/removal)
Uses	Antibiotics (penicillin produced in stationary phase)	Mycoprotein, single-cell protein

Aseptic Technique

Maintaining sterility is essential in biotechnology and microbiology to prevent contamination by unwanted organisms that could outcompete the desired organism, produce toxins, or invalidate results. Key aseptic

techniques include: sterilising equipment and media using autoclaving (121 degrees C, 15 min, at pressure); flaming inoculating loops and bottle necks; working near a Bunsen flame (creates an updraft that prevents airborne microorganisms from settling); sealing Petri dishes with tape (not completely, to allow gas exchange while preventing contamination); and disinfecting work surfaces before and after use.

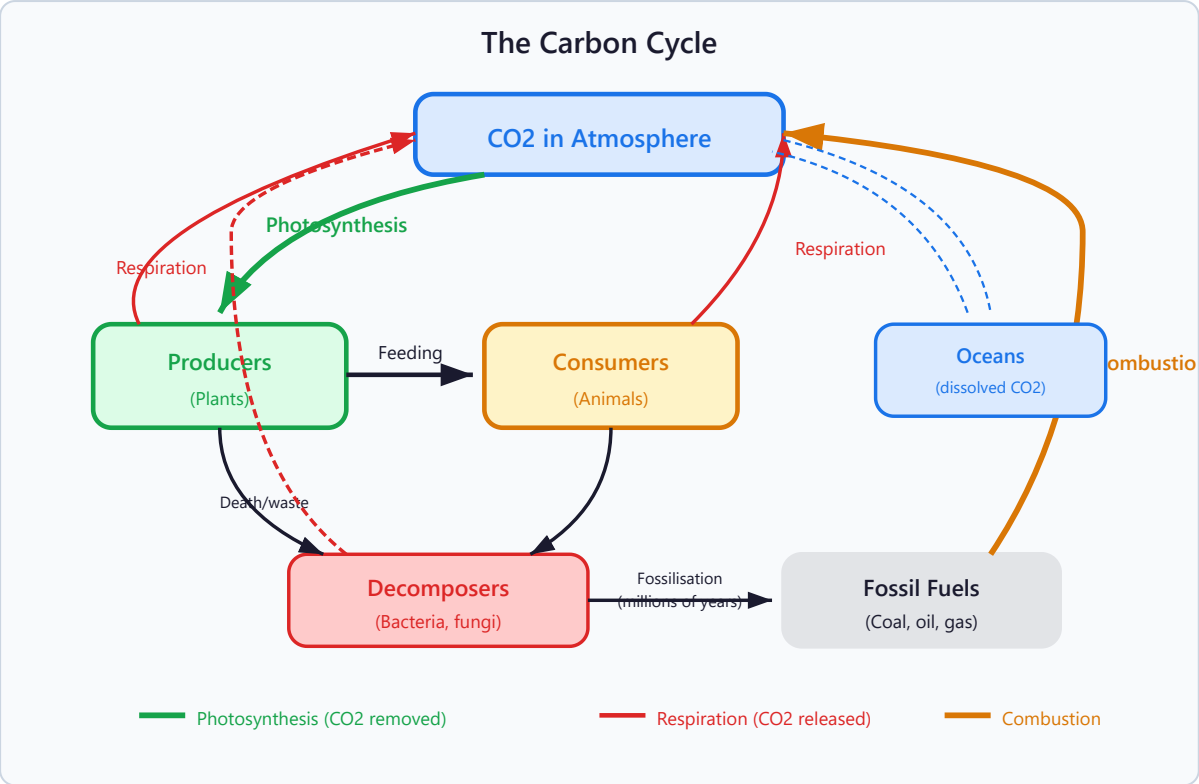
Exam Tip: When explaining biotechnology processes, always address: what organism is used and why (e.g., fast reproduction rate, simple nutritional needs, well-understood genetics), what conditions are optimised and why (temperature, pH, O_2, nutrients), how the product is extracted and purified (downstream processing), and the advantages over traditional methods.

6.4 Ecosystems

Nutrient Cycling

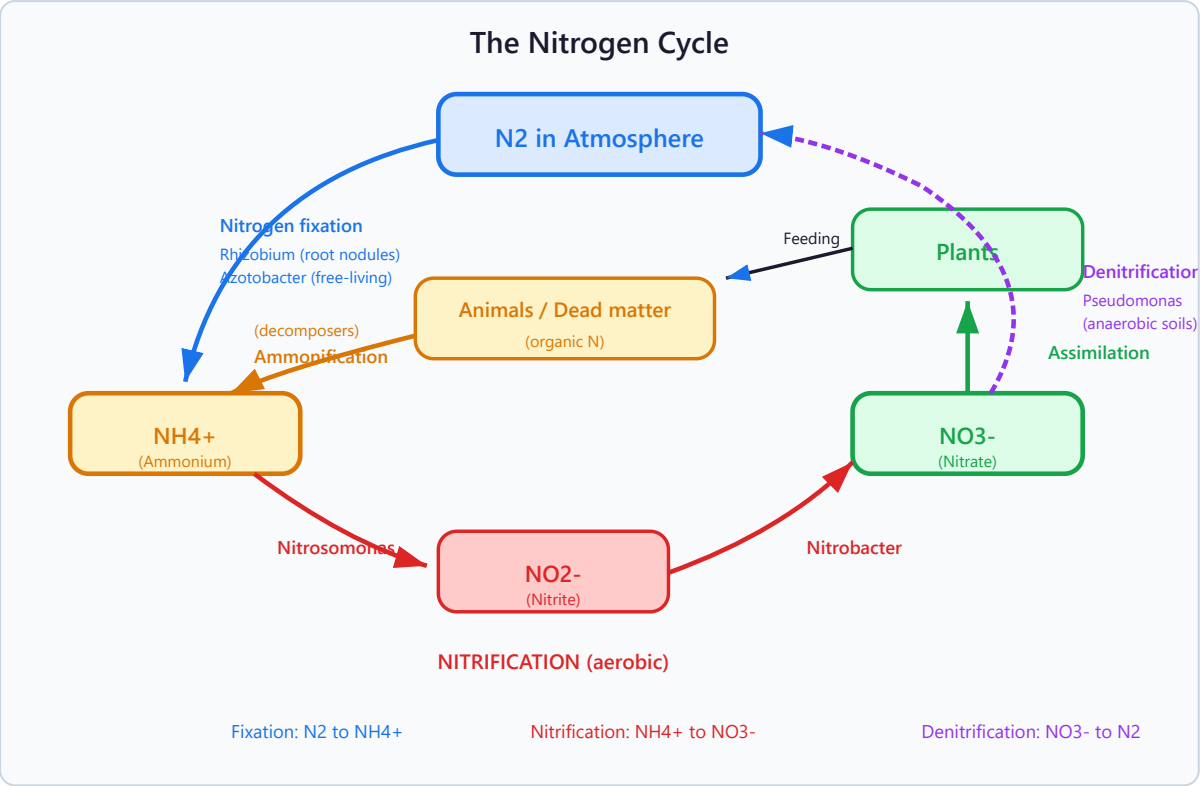
Unlike energy, which flows through ecosystems and is lost as heat, nutrients are recycled between living organisms and the abiotic environment.

The Carbon Cycle: Carbon enters living systems through photosynthesis (CO2 fixed into organic molecules). It passes through food chains and is returned to the atmosphere by respiration, decomposition, and combustion of fossil fuels. Carbon is stored in biomass, fossil fuels, oceans (dissolved CO2 and carbonates), and sedimentary rocks.



The Nitrogen Cycle: Nitrogen is essential for amino acids and nucleic acids but atmospheric N2 is unreactive. Key processes include:

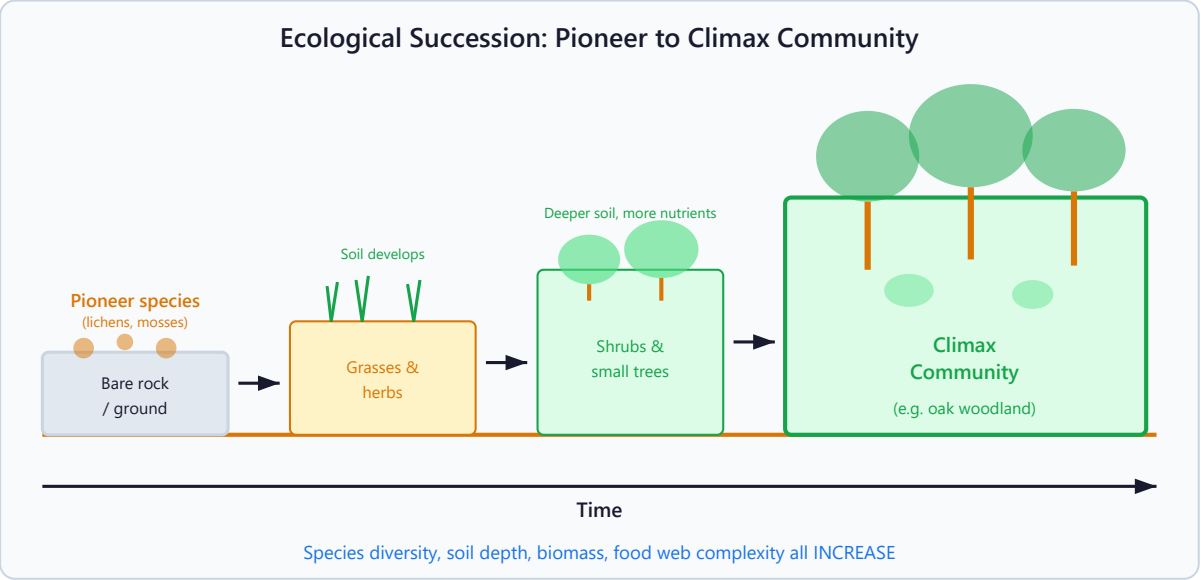
- **Nitrogen fixation:** Conversion of N2 to NH4+ by Rhizobium bacteria (in root nodules of legumes) or free-living Azotobacter, or industrially by the Haber process
- **Nitrification:** Conversion of NH4+ to NO2- (by Nitrosomonas) then NO3- (by Nitrobacter) in aerobic soil conditions
- **Assimilation:** Plants absorb NO3- or NH4+ and incorporate nitrogen into organic molecules. Animals obtain nitrogen by consuming plants
- **Ammonification:** Decomposers break down dead organic matter and excretory products, releasing NH4+ (ammonium ions)
- **Denitrification:** Anaerobic bacteria (e.g., Pseudomonas) convert NO3- back to N2 gas, returning it to the atmosphere. Occurs in waterlogged, anaerobic soils



Succession

Succession: The directional change in community composition over time, from colonisation of bare ground (primary succession) or after disturbance (secondary succession) towards a stable climax community.

Climax community: The final, stable community in a succession that is in dynamic equilibrium with the prevailing environmental conditions.



During succession, species diversity generally increases, soil depth and nutrient content increase, food webs become more complex, and biomass accumulates. Pioneer species (first colonisers) are adapted to harsh conditions and modify the environment, making it suitable for later species that outcompete them.

6.5 Conservation and Sustainability

Biodiversity and Its Importance

Biodiversity exists at three levels: genetic diversity (variation within a species), species diversity (number and relative abundance of species in a community), and ecosystem diversity (range of habitats in a region). High biodiversity confers ecosystem stability, resilience to change, and greater potential for future discoveries (e.g., medicinal compounds).

Threats to Biodiversity

- **Habitat destruction:** Deforestation, urbanisation, and agricultural expansion reduce available habitats
- **Pollution:** Eutrophication of waterways, acid rain, plastic pollution, air pollution
- **Overexploitation:** Overfishing, overhunting, unsustainable harvesting of resources
- **Climate change:** Altering temperature and rainfall patterns, shifting habitats faster than species can adapt or migrate
- **Invasive species:** Non-native species outcompeting endemic species for resources

Conservation Strategies

Conservation aims to maintain biodiversity through:

- **In situ conservation:** Protecting organisms in their natural habitats (national parks, marine reserves, SSSIs, wildlife corridors, hedgerow management)
- **Ex situ conservation:** Protecting organisms outside their natural habitat (zoos, botanical gardens, seed banks, captive breeding programmes, cryopreservation)
- **Legislation:** CITES (Convention on International Trade in Endangered Species), national wildlife protection acts, fishing quotas
- **International cooperation:** Agreements on emissions reduction, protected marine areas, biodiversity action plans

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6.6 Human Impact on the Environment

Climate Change

The enhanced greenhouse effect results from increased atmospheric concentrations of greenhouse gases (CO2, methane, N2O) due to human activities including fossil fuel combustion, deforestation, agriculture, and industrial processes. These gases absorb and re-radiate infrared radiation, increasing global temperatures.

Consequences include rising sea levels (thermal expansion and ice melt), more frequent extreme weather events, shifting biomes, coral bleaching, changes in species distributions, and threats to food security.

Eutrophication

Excess fertiliser (nitrates and phosphates) from agricultural runoff enters waterways, promoting rapid algal growth (algal bloom). The algae block light, killing aquatic plants beneath. When algae die, decomposing bacteria increase in number and deplete dissolved oxygen through aerobic respiration. This creates anaerobic conditions, killing fish and other aquatic organisms.

Sustainability

Sustainability: Meeting the needs of the present generation without compromising the ability of future generations to meet their own needs. In ecology, this means using resources at a rate that allows them to be replenished naturally.

Sustainable practices include managed forestry (replanting, selective felling, coppicing), sustainable fishing (quotas, mesh sizes, exclusion zones, rotation), organic farming (reducing chemical inputs), renewable energy sources, and reducing waste through recycling and circular economy approaches.

Exam Tip: When discussing human impacts, always consider both ecological and economic perspectives. Conservation strategies must often balance biodiversity protection with the livelihoods and needs of local communities. Examiners reward answers that demonstrate understanding of these competing interests and the need for compromise.

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